

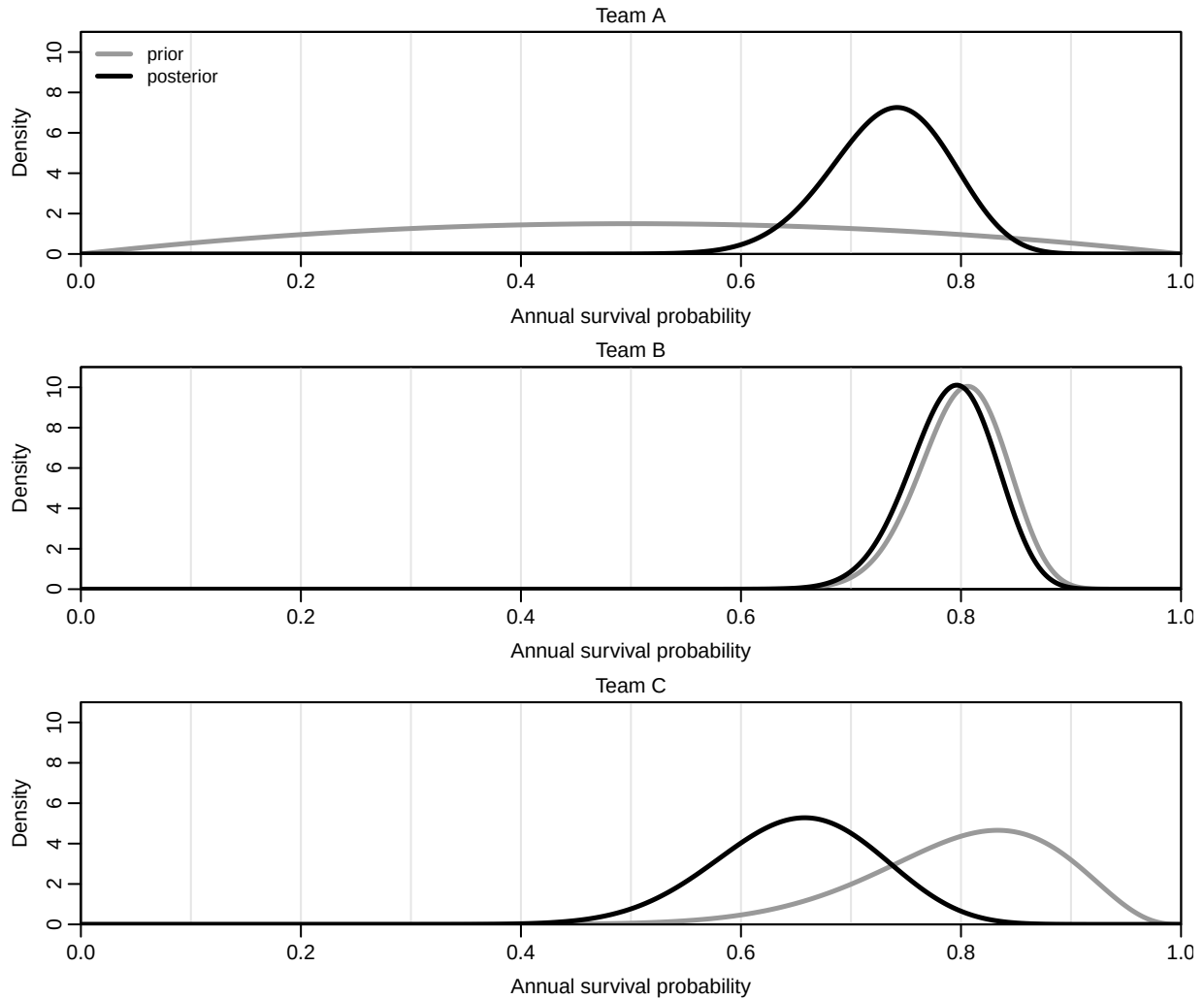
NRES 746, Week 8, Lecture 1: How Much Did the Prior Matter?

Name: _____

Instructions: Work alone on Part 1 (about 4 minutes), then compare with a neighbor for Part 2. Keep laptops closed.

The studies

Three research teams estimate the annual survival probability of adult desert tortoises at three different sites, using radio-collared animals. Each team used a prior based on earlier work and a different number of collared tortoises. Each panel shows one team's prior (grey) and posterior (black). The likelihoods are not shown.



Part 1: Read the panels (individual)

1a. Rank the three teams from most to least influenced by their prior. Give a one-line reason for each.

1b. Sketch the shape of Team C's likelihood on its panel (the height doesn't matter). Where does it peak, and is it wider or narrower than the prior?

Part 2: How much data? (with a neighbor)

2a. Which team collared the most tortoises, and which collared the fewest? Team B has the narrowest posterior of the three. Does that mean Team B had the most data?
